

Jewel Haque

🏠 Room #149, Petrophysical Lab., Dept. of PMRE, BUET, Dhaka-1000

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in [linkedin.com/in/jewelhaque15](https://www.linkedin.com/in/jewelhaque15) | 🌐 sites.google.com/view/jewelhaque15 | 📺 [Mechanical Munshi](#)

Research Profile

- Experimental & computational researcher with a multidisciplinary academic & research background focused on **reservoir & geomechanics simulation, Instrumentation, CFD, heat & mass transfer, molecular dynamics** of energy systems.
- Practical experience in experimental research in energy & sensor development along with computational modeling using **CMG GEM/STARS, ANSYS Fluent, LAMMPS, Python, MATLAB**

Research & Publications

- **Conference Paper (Accepted – Poster, ICChE 2026):** Jewel Haque et al. “Operational Geomechanical Assessment of Caprock Integrity Under Cyclic CO₂ Injection in a Depleted Gas Reservoir Using Coupled Flow–Geomechanics Simulation with CMG GEM.” *8th International Conference on Chemical Engineering (ICChE 2026)*, Paper ID 212.
- **Journal Article:** Jewel Haque, Md. Ibrahim Al Imran, Khondkar Siddique-e-Rabbani. “Sensitivity Optimization of an Optical Pressure Sensor-Based Low-Cost Dynamic Pedograph for Improved Foot Pressure Analysis.” *Int. J. of Biomedical Engineering and Technology*
<https://doi.org/10.1504/IJBET.2026.151950>
- **Book (2024):** *MEkanika: A Consolidated Hand-note on Mechanical Engineering Fundamentals*.

Education

- **M.Sc.Eng. (pursuing, April 2024), Petroleum & Mineral Resources Engineering**, Bangladesh University of Engineering and Technology (BUET).
Focus: Fluid flow in porous media, heat transport, oil-gas properties, petro-geology, energy policy and economics, Drilling and reservoir engineering with simulation.
- **M.S., Biomedical Physics & Technology**, University of Dhaka.
Thesis: Sensitivity Optimization of an Optical Sensor-Based Dynamic Pedograph.
Enhanced optical pressure sensor sensitivity through calibration protocols and signal processing, enabling accurate dynamic foot pressure analysis for clinical applications. CGPA: 3.05/4.00 (2024)
- **B.Sc.Eng., Mechanical Engineering**, Bangladesh University of Engineering and Technology (BUET).
Thesis: Study on Fruits and Vegetables Drying with a Vacuum Drying Machine.
Built a low-cost vacuum dryer using locally sourced materials, then analyzed drying kinetics and crop quality to develop an energy-efficient preservation method CGPA: 2.85/4.00 (2021)

Professional Experience

1. Research Experience

1.1 Assistant Experimental Engineer, Petrophysical & Analytical Lab, PMRE, BUET Aug 2023 – Present

- Conduct experimental studies on petrophysical properties of reservoir rock-fluid and petroleum products.

- Develop and maintain lab protocols (ASTM-compliant testing), prepare lab manuals and instruments, and support departmental and consulting R&D projects.
- Run reservoir and coupled flow–geomechanics simulations in **CMG GEM/STARS** (CO₂ storage (CCUS), caprock integrity, cyclic injection)
- Support M.Sc. and B.Sc. students as lab personnel — assisting with experimental, data analysis, and simulation work.

1.2 Research Engineer (Part-time), BiBeat Ltd.

Jan 2023 – Aug 2024

- Biomedical device and signal processing prototyping in collaboration with cross-functional engineering teams.

2. Industrial & Technical Experience

2.1 Mechanical Design Engineer, Industrial Technical Services

Nov 2022 – Dec 2023

- Designed and manufactured mechanical parts using CAD, precision machining, and heat treatment, including a waste-heat recovery system to fix local factory problems and improve efficiency.

2.2 Management Trainee Officer, VitaCan Industries Ltd.

Mar 2022 – Nov 2022

- Supervised mechanical maintenance tasks, coordinated automated can-line installation, PLC and other microcontroller-based tasks, and managed operational troubleshooting.

2.3 Trainee Engineer - Production & Development, Habiganj Industrial Park (HIP), PRAN–RFL Group

March 2021 – Nov. 2021

- Gained hands-on experience with industrial machinery (compressors, boilers, chillers, ETP), quality control, preventive maintenance, and plant scheduling.

3 Teaching & Outreach

3.1 Engineering Instructor, International Maritime Training Academy (IMTA)

Dec 2022 – Aug 2023

- Taught Engineering Mechanics, Heat Engineering, Naval Architecture, and Electrical Technology; developed teaching materials, problem sets, and competency assessments.

3.2 Content Creator, YouTube — “[Mechanical Munshi](#)”

2019 – Present

- Produced 330+ lectures on core mechanical engineering topics (Fluid Mechanics, Thermodynamics, Solid Mechanics, Heat Transfer, Engineering Drawing, etc).

3.3 Math. Instructor, 10 Minute School (Online Educational Platform)

2019–2022

- Created higher secondary-level math content focusing on conceptual clarity, along with numerical problem-solving ability for mass-scale learners (engineering university admission seekers)

Technical Skills & Certifications

- **Reservoir & Geomechanics Simulation:** CMG GEM (coupled flow–geomechanics, CO₂ storage, cyclic injection modelling), CMG STARS.
- **CFD & Simulation:** ANSYS Fluent, COMSOL, SimScale.

- **Molecular Dynamics:** LAMMPS (basic to intermediate) and OVITO, VMD for visualisation tools
- **Programming & HPC:** Python (NumPy, SciPy, Matplotlib), MATLAB, C/C++ (basic), Linux, MPI fundamentals.
- **Design:** SolidWorks, AutoCAD, AutoCAD Plant 3D, MasterCAM.
- **Languages:** Bengali (Native), English (Fluent).
- **Certifications:**
 - Method Development and Sample Analysis by GC, GC-MS & HPLC (TICI, 2024)
 - [Linear Regression with NumPy and Python](#) (Coursera, 2022)
 - [Computational Fluid Mechanics - Airflow Around a Spoiler](#) (Coursera, 2022)
 - [Programming for Everybody \(Getting Started with Python\)](#) (Coursera, 2022)
 - Energy Auditor Certification (SREDA, 2025, attended)

Affiliations

- Affiliated Member, Institution of Mechanical Engineers (IMechE). Membership No.: 80536530
- President, SPE Student Chapter, Bangladesh University of Eng & Tech Chapter; Member ID: 5945894

References

- **Dr. Shahriar Mahmud**
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